PBL Portal Content - Devkinandan Kumawat

Smart Hospital Management System (SmartCare) | Team Nobita (PBL2627-AI&DS-A-031;) | Mentor: Er. Ram Babu Buri

Member	Devkinandan Kumawat (24E1ARADM30P042)
Module	Module 5 - Billing, Admission, Reports and Feedback
Code owned	Bill, Payment, Admission, Room, Feedback, Notification models and DAOs, BillingService, AdmissionService, BillingServlet, AdmissionServlet, invoice JSPs, bills JSPs, admit_patient, admission_list, rooms, admin reports and billing JSPs, patient feedback.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

How to use this file

1) Daily Work Log tab: pick the Date, type Hours Worked, keep Status as shown, paste the description, click Submit Today's Work. 2) Weekly Report tab: choose the Week, paste the GitHub link, then paste the three boxes (Work Completed, Problems Faced, Plan for Next Week) from Part A. 3) Part A (weekly) comes first, Part B has the day-wise table. Submit your own entries only, from your own login.

PART A - Weekly Reports (13 weeks)

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Joined team of five and discussed the project idea together. (2) Studied how hospital billing and admission work. (3) Wrote the abstract section on billing, admission and reports. (4) Discussed technology options with the team and guide. (5) Checked the final abstract and helped with submission.
Problems Faced	Agreeing on a single project scope. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Listed requirements for bill generation and payments. (2) Listed requirements for room allocation and admission. (3) Listed requirements for admin reports and revenue summary.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Listed requirements for bill generation and payments. (2) Listed requirements for room allocation and admission. (3) Listed requirements for admin reports and revenue summary. (4) Listed requirements for patient feedback and notifications. (5) Compiled these into the SRS and reviewed with the team.
Problems Faced	Unclear billing rules at the start. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for billing and admission. (2) Designed the class diagram for Bill and Payment. (3) Designed the class diagram for Admission and Room.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Drew the use-case diagram for billing and admission. (2) Designed the class diagram for Bill and Payment. (3) Designed the class diagram for Admission and Room. (4) Drew the sequence diagram for generating an invoice. (5) Merged the module UML into the team design document.
Problems Faced	Relation between admission and room. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed the billing and payments tables. (2) Designed the rooms and admissions tables. (3) Designed the feedback and notifications tables.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Designed the billing and payments tables. (2) Designed the rooms and admissions tables. (3) Designed the feedback and notifications tables. (4) Set up local development environment and cloned the repo. (5) Drew mock-ups for invoice, admission and reports screens.
Problems Faced	Choosing data types for amounts. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote Bill and Payment model classes. (2) Wrote Admission and Room model classes. (3) Wrote Feedback and Notification models.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote Bill and Payment model classes. (2) Wrote Admission and Room model classes. (3) Wrote Feedback and Notification models. (4) Implemented BillingDAO with bill and payment queries. (5) Implemented AdmissionDAO and RoomDAO with availability queries.
Problems Faced	Decimal rounding in totals. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Built BillingService to calculate totals and taxes. (2) Built AdmissionService for admit, transfer and discharge logic. (3) Implemented BillingServlet for bill generation and payment.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Built BillingService to calculate totals and taxes. (2) Built AdmissionService for admit, transfer and discharge logic. (3) Implemented BillingServlet for bill generation and payment. (4) Implemented AdmissionServlet for admit and discharge actions. (5) Added room availability updates on admission and discharge.
Problems Faced	Room status not updated on discharge. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created the invoice.jsp with printable layout. (2) Created receptionist billing.jsp and generate_bill.jsp. (3) Created common admit_patient.jsp and admission_list.jsp.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Created the invoice.jsp with printable layout. (2) Created receptionist billing.jsp and generate_bill.jsp. (3) Created common admit_patient.jsp and admission_list.jsp. (4) Created admin rooms.jsp and receptionist rooms.jsp. (5) Created patient bills.jsp and feedback.jsp.
Problems Faced	Invoice print layout broke on paper size. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Added validation for amount, payment mode and dates. (2) Handled exceptions for occupied room and duplicate payment. (3) Prevented overpayment greater than the bill total.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Added validation for amount, payment mode and dates. (2) Handled exceptions for occupied room and duplicate payment. (3) Prevented overpayment greater than the bill total. (4) Restricted billing pages by role. (5) Tested validation messages on all forms.
Problems Faced	Overpayment case was missed. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Completed the admin reports.jsp with revenue and admission figures. (2) Added payment status filters to the bills list. (3) Added the notification entries on bill generation.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Completed the admin reports.jsp with revenue and admission figures. (2) Added payment status filters to the bills list. (3) Added the notification entries on bill generation. (4) Reviewed code and added comments in service classes. (5) Pushed module code to GitHub and updated the sheet.
Problems Faced	Report query was slow with many joins. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Linked bills to appointments and prescriptions from other modules. (2) Linked admission with patient and room lists. (3) Connected the reports page to data from all modules.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Linked bills to appointments and prescriptions from other modules. (2) Linked admission with patient and room lists. (3) Connected the reports page to data from all modules. (4) Fixed total mismatch after merging charges. (5) Ran full flow from admission to discharge and invoice.
Problems Faced	Totals mismatched after merging charges. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote test cases for billing and payments. (2) Tested tax and total calculations with samples. (3) Tested admission, room status and discharge.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote test cases for billing and payments. (2) Tested tax and total calculations with samples. (3) Tested admission, room status and discharge. (4) Tested report totals against database values. (5) Fixed issues and retested.
Problems Faced	Rounding difference in a test case. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Checked billing, admission and report pages on live site. (2) Fixed decimal formatting on the cloud database. (3) Corrected a missing room status default in seed data.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Checked billing, admission and report pages on live site. (2) Fixed decimal formatting on the cloud database. (3) Corrected a missing room status default in seed data. (4) Retested invoice printing on the live server. (5) Noted deployment fixes for the report.
Problems Faced	Decimal formatting changed on cloud DB. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the billing, admission and reports section of the report. (2) Added invoice and report screenshots. (3) Prepared slides on billing and admission workflow.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote the billing, admission and reports section of the report. (2) Added invoice and report screenshots. (3) Prepared slides on billing and admission workflow. (4) Rehearsed demo of invoice generation and discharge. (5) Prepared viva answers on billing logic and room status.
Problems Faced	Collecting output screenshots. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

PART B - Daily Work Log (Monday to Friday, 65 entries)

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description
06-07-2026 (Mon)	1.75	Completed	Joined team of five and discussed the project idea together.
07-07-2026 (Tue)	2	Completed	Studied how hospital billing and admission work.
08-07-2026 (Wed)	1.75	Completed	Wrote the abstract section on billing, admission and reports.
09-07-2026 (Thu)	1.5	Completed	Discussed technology options with the team and guide.
10-07-2026 (Fri)	1.75	Completed	Checked the final abstract and helped with submission.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description
13-07-2026 (Mon)	2	Completed	Listed requirements for bill generation and payments.
14-07-2026 (Tue)	1.5	Completed	Listed requirements for room allocation and admission.
15-07-2026 (Wed)	2	Completed	Listed requirements for admin reports and revenue summary.
16-07-2026 (Thu)	1.75	Completed	Listed requirements for patient feedback and notifications.
17-07-2026 (Fri)	1.75	Completed	Compiled these into the SRS and reviewed with the team.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description
20-07-2026 (Mon)	1.75	Completed	Drew the use-case diagram for billing and admission.
21-07-2026 (Tue)	1.5	Completed	Designed the class diagram for Bill and Payment.
22-07-2026 (Wed)	2	Completed	Designed the class diagram for Admission and Room.
23-07-2026 (Thu)	2	Completed	Drew the sequence diagram for generating an invoice.
24-07-2026 (Fri)	2	Completed	Merged the module UML into the team design document.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description
27-07-2026 (Mon)	1.75	Completed	Designed the billing and payments tables.
28-07-2026 (Tue)	2	Completed	Designed the rooms and admissions tables.
29-07-2026 (Wed)	2	Completed	Designed the feedback and notifications tables.
30-07-2026 (Thu)	1.75	Completed	Set up local development environment and cloned the repo.
31-07-2026 (Fri)	2	Completed	Drew mock-ups for invoice, admission and reports screens.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description
03-08-2026 (Mon)	2	Completed	Wrote Bill and Payment model classes.
04-08-2026 (Tue)	1.5	Completed	Wrote Admission and Room model classes.
05-08-2026 (Wed)	1.5	Completed	Wrote Feedback and Notification models.
06-08-2026 (Thu)	1.75	Completed	Implemented BillingDAO with bill and payment queries.
07-08-2026 (Fri)	2	Completed	Implemented AdmissionDAO and RoomDAO with availability queries.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description
10-08-2026 (Mon)	1.75	Completed	Built BillingService to calculate totals and taxes.
11-08-2026 (Tue)	1.75	Completed	Built AdmissionService for admit, transfer and discharge logic.
12-08-2026 (Wed)	2	Completed	Implemented BillingServlet for bill generation and payment.
13-08-2026 (Thu)	2	Completed	Implemented AdmissionServlet for admit and discharge actions.
14-08-2026 (Fri)	1.5	Completed	Added room availability updates on admission and discharge.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description
17-08-2026 (Mon)	2	Completed	Created the invoice.jsp with printable layout.
18-08-2026 (Tue)	1.5	Completed	Created receptionist billing.jsp and generate_bill.jsp.
19-08-2026 (Wed)	1.5	Completed	Created common admit_patient.jsp and admission_list.jsp.
20-08-2026 (Thu)	1.75	Completed	Created admin rooms.jsp and receptionist rooms.jsp.
21-08-2026 (Fri)	2	Completed	Created patient bills.jsp and feedback.jsp.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description
24-08-2026 (Mon)	2	Completed	Added validation for amount, payment mode and dates.
25-08-2026 (Tue)	1.75	Completed	Handled exceptions for occupied room and duplicate payment.
26-08-2026 (Wed)	1.75	Completed	Prevented overpayment greater than the bill total.
27-08-2026 (Thu)	1.75	Completed	Restricted billing pages by role.
28-08-2026 (Fri)	1.75	Completed	Tested validation messages on all forms.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description
31-08-2026 (Mon)	1.75	Completed	Completed the admin reports.jsp with revenue and admission figures.
01-09-2026 (Tue)	1.5	Completed	Added payment status filters to the bills list.
02-09-2026 (Wed)	1.5	Completed	Added the notification entries on bill generation.
03-09-2026 (Thu)	1.75	Completed	Reviewed code and added comments in service classes.
04-09-2026 (Fri)	2	Completed	Pushed module code to GitHub and updated the sheet.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description
07-09-2026 (Mon)	2	Completed	Linked bills to appointments and prescriptions from other modules.
08-09-2026 (Tue)	1.75	Completed	Linked admission with patient and room lists.
09-09-2026 (Wed)	1.5	Completed	Connected the reports page to data from all modules.
10-09-2026 (Thu)	2	Completed	Fixed total mismatch after merging charges.
11-09-2026 (Fri)	2	Completed	Ran full flow from admission to discharge and invoice.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description
14-09-2026 (Mon)	2	Completed	Wrote test cases for billing and payments.
15-09-2026 (Tue)	1.5	Completed	Tested tax and total calculations with samples.
16-09-2026 (Wed)	1.5	Completed	Tested admission, room status and discharge.
17-09-2026 (Thu)	2	Completed	Tested report totals against database values.
18-09-2026 (Fri)	1.5	Completed	Fixed issues and retested.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description
21-09-2026 (Mon)	1.75	Completed	Checked billing, admission and report pages on live site.
22-09-2026 (Tue)	1.75	Completed	Fixed decimal formatting on the cloud database.
23-09-2026 (Wed)	2	Completed	Corrected a missing room status default in seed data.
24-09-2026 (Thu)	2	Completed	Retested invoice printing on the live server.
25-09-2026 (Fri)	1.75	Completed	Noted deployment fixes for the report.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description
28-09-2026 (Mon)	1.75	Completed	Wrote the billing, admission and reports section of the report.
29-09-2026 (Tue)	2	Completed	Added invoice and report screenshots.
30-09-2026 (Wed)	2	Completed	Prepared slides on billing and admission workflow.
01-10-2026 (Thu)	1.75	Completed	Rehearsed demo of invoice generation and discharge.
02-10-2026 (Fri)	1.75	Completed	Prepared viva answers on billing logic and room status.

Total hours in this log: 116.75 over 65 days.